LS-CAT Management Board Meeting

December 12, 2014

Agenda

Call In Number: 1-888-850-4523
Passcode: 190580

http://conf1.ls-cat.org
http://conf2.ls-cat.org

9 AM - 11 AM CST (10AM - Noon EST)

Approval of Minutes from Last Meeting

Eiger 9M Detector

APS SAC Review of LS-CAT

Long term planning / Renewal / Financials

LS-CAT Management Board Chair

Beamline Status

- Stability problems with 21-ID-D monochromator
- The “smart” magnet
- New remote access interface starts 2015-1

Next Meeting

New Business

Eiger detector

An Eiger 9M detector was ordered this summer for delivery in October 2015.

We have the option to upgrade it to an Eiger 16M if we do so before March 2015 (and pay the difference in price)

Detector	Eiger X 16M	Eiger X 9M	Pilatus3 X 6M	Pilatus3 S 6M	Rayonix MX 300 HS	ADSC HF-4M
Active Area	311 x 328	233 x 245	424 x 435	424 x 435	300 x 300	340 x 310
Gaps	6.60%	6.30%	8.50%	8.50%	0.1 %	8.5 %
# Edge Pixels	98,816 (0.5%)	74,112 (0.7%)	81,840 (1.3%)	81,840 (1.3%)	245,760 (1.7%)	81,920 (2.0%)
Nominal Pixel Size	75 x 75	75 x 75	172 x 172	172 x 172	2 x 2 bin: 78 x 78	150 x 150
Nominal # Pixels	18,139,650	10,166,590	6,224,001	6,224,001	14,745,600	4,194,304
Maximum Speed (Hz)	187	238	100	25	10	200
Deadtime (ms)	0.003	0.003	0.95	2	1	0.001
Sensor	450 µm Si	450 µm Si	450 µm Si	450 µm Si	40 µm Gd2O2S	500 µm Si
Read Noise	0	0	0	0	8 e/pixel (fast) 4 e/pixel (slow)	0 (counting) 0.2 (integrating)
Point Spread Function FWHM (µm)	75	75	172	172	100	<150
Photons/Second per pixel @ 12 KeV	2E+06	2E+06	1E+07	1E+07	5E4 (fast mode) 1E5 (slow mode)	1E5 (counting w/ 1% loss) 1E8 (integrating)

APS SAC Review of LS-CAT November 4, 2014

8:00	8:30	Executive Session (APCF Conference Room)	
8:30	8:45	Life Sciences Capabilities at the APS	Robert Fischetti
8:45	8:55	(Short walk to 436C)	
8:55	10:00	LS-CAT Science Overview, History, and Organization LS-CAT Design and User Program	Alfonso Mondragón Keith Brister
10:00	10:45	Biological methane oxidation	Amy Rosenzweig
10:45	11:00	Coffee Break	
11:00	11:30	LS-CAT Enabling Technologies	Keith Brister
11:30	12:30	The Future of LS-CAT: Discussion	Alfonso and Keith
12:30	1:30	Working Lunch with LS-CAT staff (without management)	
1:30	2:15	BNP Science	Gayle Woloschak
2:15	3:00	Structural mechanisms of DNA base excision and replication repair	Brandt Eichman
3:00	3:30	Coffee Break and Beamline Tour	
3:30	5:00	Executive Session	
5:00		Adjourn	

Committee Members

Britt Hedman, Chair, SSRL
Paul Adams, LBNL
Christopher Hill, University of Utah Medical School
Lisa Miller, BNL

There were two parts of the of the review: a written summary and an oral presentation. A link to the meeting materials has been provided to you.

Evaluation Criteria

1. Scientific Mission and Impact
2. Beamline Productivity
3. Leveraging of specific APS characteristics
4. Effectiveness of beam time use
5. General User program quality
6. Adequacy and reliability of funding
7. Future Plan
8. Overall
9. Recommended Actions including recommendation to APS for continuation, a re-review before five years, or termination/decommissioning.

At the closeout the committee sounded impressed by all 7 of the specific areas above and will be recommending that we continue. We are hopeful that a written report will be available to us in January.

The Future of LS-CAT

(as presented to the APS SAC Review Committee)

Near term Future of LS-CAT

The current 15 year LS-CAT agreement between members expires February 28, 2017.

We have 7 members owning a total of 5.5 shares of the CAT. WSU has indicated that it would like to rejoin at which point we would have 8 members with a total of 6 shares.

At this point a 5 year extension of the existing agreement seems likely. The biggest sticking points are:

1) The timing of the APS-U dark period.

2) The desire to maintain constant dues (plus inflation) regardless of the number of other members.

We recognize the desirability of increasing the number of LS-CAT shares to help with the capital costs of upgrades and to increase the depth of the available staff.

Near term risks and opportunities

Risks

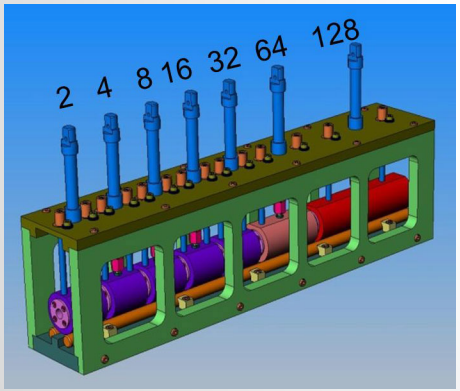
1) Not enough members sign on to the extension
2) Delays in signing the contract cause staff to lose confidence
3) Capital improvement needs increase faster than our capital budget

Opportunities

1) New members join bringing new capital and operations funds
2) Existing members demand an increase in the budget to enhance LS-CAT activities (greater staff coverage, greater BNP usage, commissioning of 21-ID-E)

LS-CAT Near Future: F/G Optics

Currently our F and G stations lose about 20% intensity due to a mismatch between the Be lens diameter and the beam width. By using a larger diameter lens system (that was not available when we first installed CRL's) and, perhaps, an arrangement such as a "transfocator" one should be able to both increase the flux and/or reduce the beam size to match the crystal size.



ESRF Transfocator
Snigirev, AA, 25 August 2011, SPIE Newsroom. DOI: 10.1117/2.1201108.003688

LS-CAT Near Future: Detectors & Data Handling

- To remain competitive we will be looking to replace our remaining CCD detectors with pixel array detectors. The details will depend on what the market has to offer and, of course, the availability of funds. Our first Eiger detector is planned to be delivered in the Fall of 2015.



- Our computing infrastructure will need an overhaul soon. We will be increasing the spinning storage capacity from it's current 54 TB size. Our current storage architecture is quite flexible and we may implement some increase in capacity by using a mixture of solid state drives and more conventional but less expensive high capacity drives. As with anything digital storage related, we will wait as long as possible before committing to an exact configuration. The first round of upgrades will occur in September 2015 before the Eiger detector is commissioned.

- As the price of implementing 10G connections decreases and our need for such connections increases we will be implementing an expansion of our 10G internal network. Note that currently we have 4G connections for critical links.

LS-CAT Near Future: Robotics

Our current sample changers cause us the greatest operational problems. One possible solution may be to adapt an “ALS” style dewar (of which there are several well engineered examples nearby) to the cryotong gripper of the CATS robot.

Reliability	Staff Involvement	Overnight Personnel
90%	Constant	On-Site
99%	Hourly	On-Site
99.9%	Daily	On-Site
99.99%	Weekly	**On-Call**

Current operations are somewhere between 99% and 99.9% (when running well).

Desired reliability is here but is unlikely be be reached using the CATS SPINE gripper.

LS-CAT Near Term: 21-ID-E

- While this station has always been on our roadmap, a pressing need to turn it over for user operations has never materialized. Partially this is due to the economic slowdown starting in 2008 (about when the other 3 stations were fully commissioned with the existing instrumentation).

- The current mission for 21-ID-E is to support off-line software development for our remote-access service. Additionally, it provides a supply of swappable spare parts to bring the other stations back to life after hardware failures.

- When the demand for this station requires we do so we will commission the 21-ID-E monochromator and bring the station on-line.

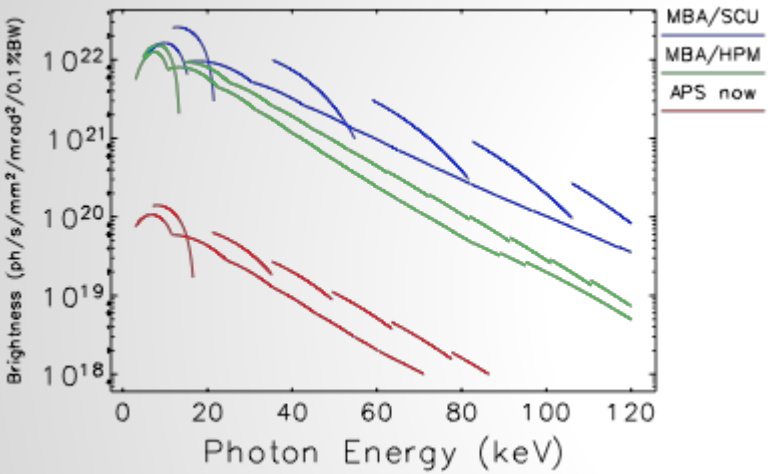
- Note: It is not practical to place a nanoprobe in the station until the MBA is implemented.

LS-CAT Near Term: 21-ID-D Mono

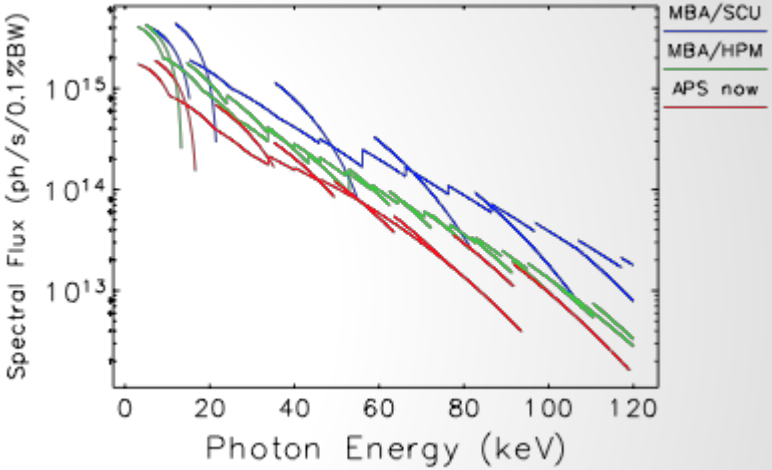
The bionanoprobe would likely benefit by the implementation of a multilayer monochromator to increase the bandpass (Si 111 resolution is not needed here). For that matter, micro-crystal MX experiments using fine-phi slicing available with the Eiger detector planned for 2015 would also likely benefit.

Likely APS MBA Upgrade

Brightness



Flux



Bottom Line: If we do not upgrade LS-CAT users will see a factor of 2 to 4 improvement. If we do upgrade they will see a factor of 100.

The APS upgrade will require a 12 to 18 month down period currently thought to be ~~around 2019~~ in the 2020's.

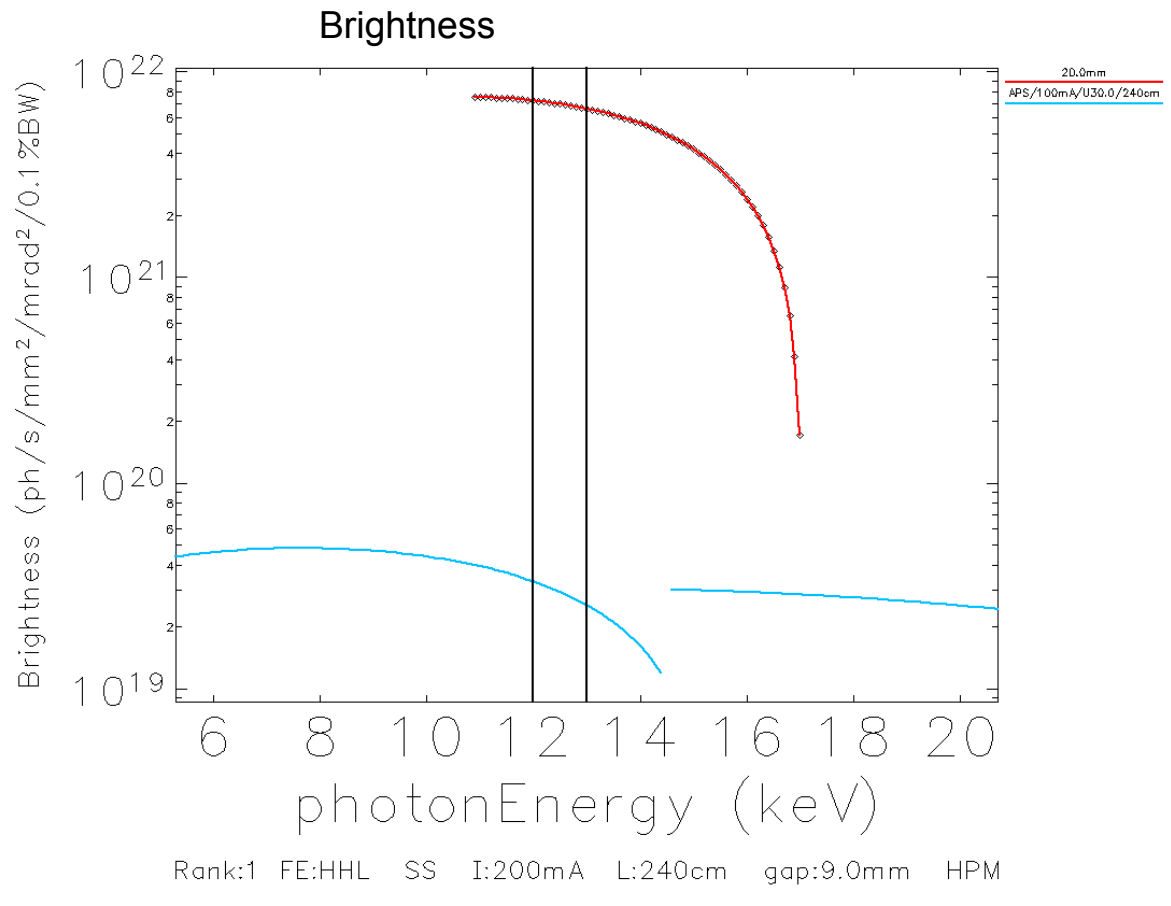
The APS Upgrade Base Scope

While it is too early to know for sure what is in the base scope of the APS upgrade, what has been talked about includes:

1) New frontends with “GRID” style BPMS
2) Optimized undulators (possibly superconducting undulators)
3) A down period of no more than one year

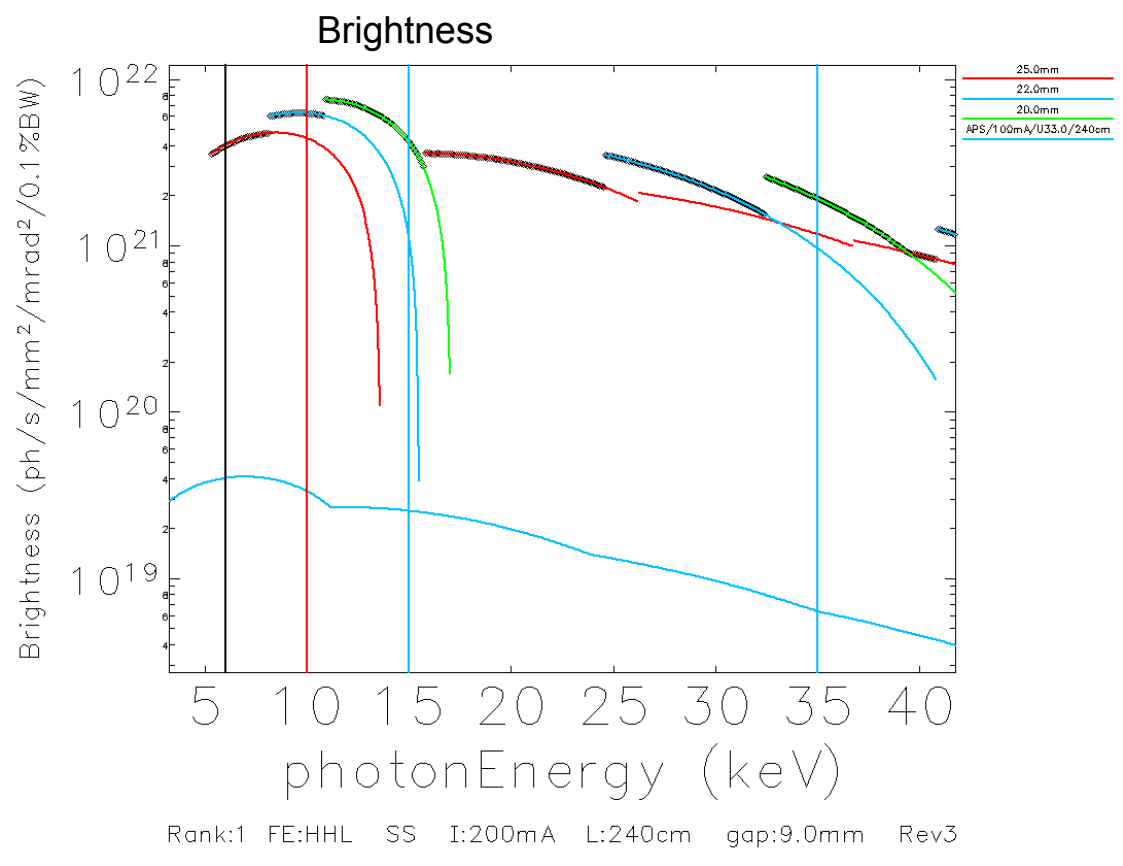
We fully support inclusion of these items as they would mean our user programs would see immense benefits from the APS upgrade.

Brightness Optimized Undulator for 21-ID-E/F/G



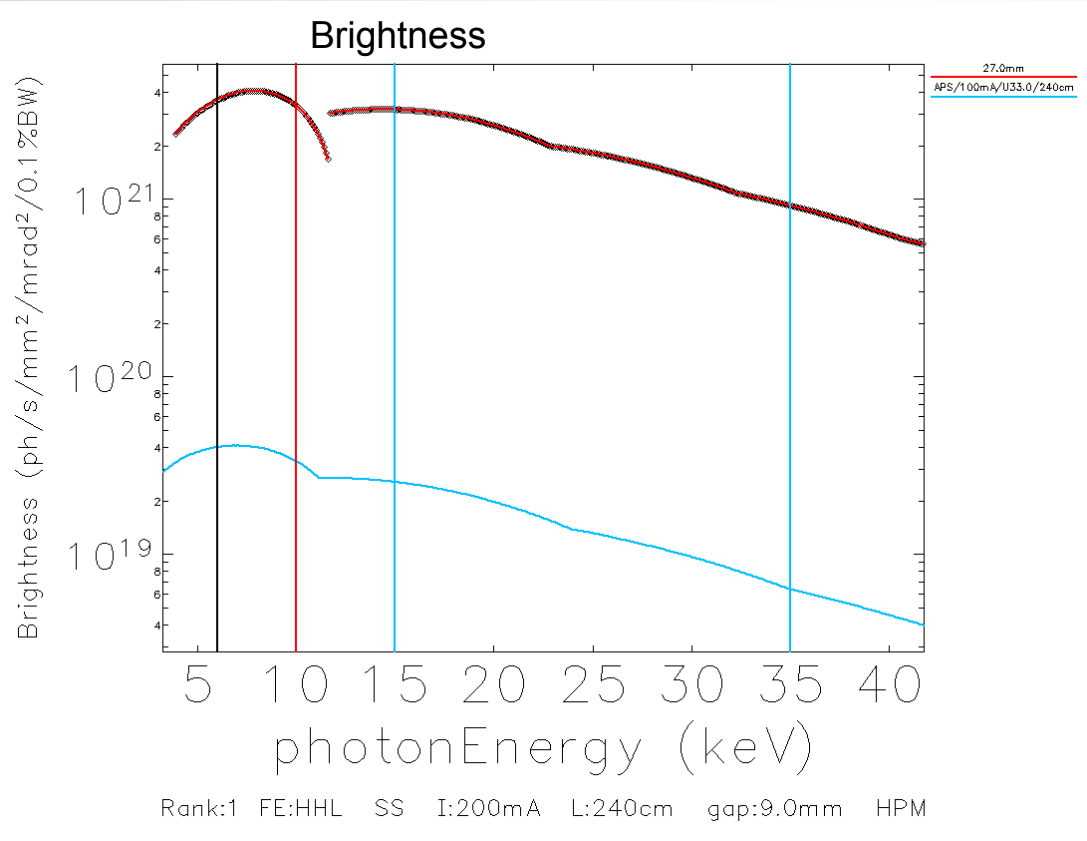
This is for a 20 mm period device.

Brightness Optimized Undulators for 21-ID-D



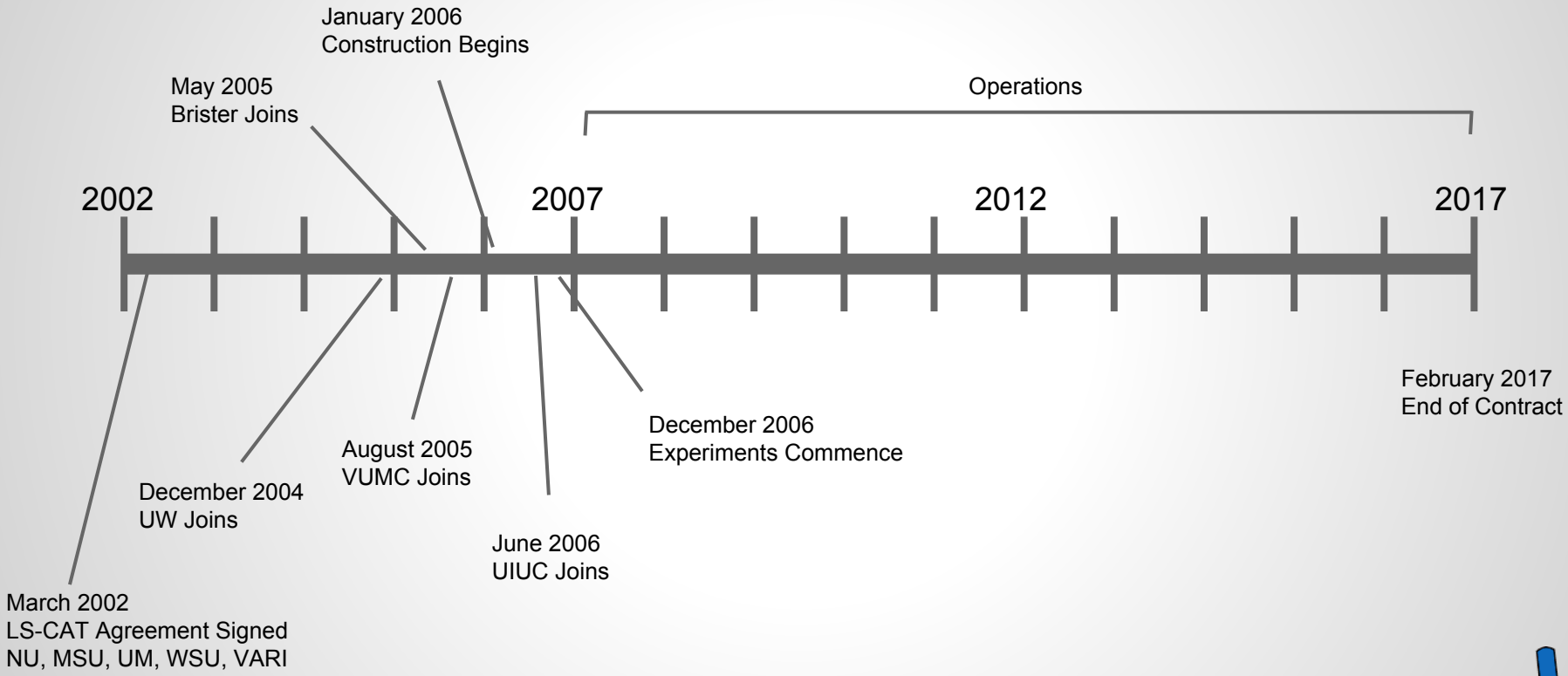
This is for a triple revolver. One undulator with 3 different sets of magnetic structures. Shown are 25mm (red), 22mm (blue), and 20mm.

Brightness Optimized Undulator for 21-ID-D

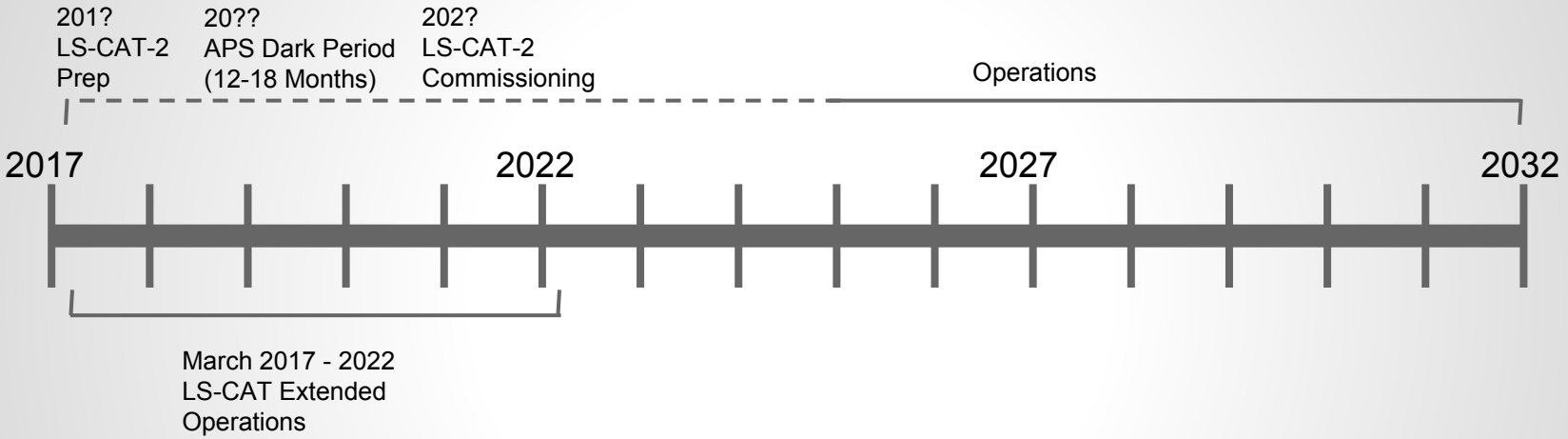


This is for a single 27 mm period device.

LS-CAT Timeline



LS-CAT-2 Timeline (Draft)



LS-CAT-2 Capital Costs (if APS covers IDs and FEs)

Item	Cost Each	Cost Uncertainty	Quantity	Total	Total Uncertainty
Station Modifications	0.3M	0.2M	1	0.3M	0.2M
Detectors	2.0M	0.5M	4	8.0M	2.0M
Focusing	0.5M	0.2M	4	2.0M	0.4M
Monochromators	0.5M	0.3M	4	2.0M	1.2M
Goniometry	0.6M	0.2M	4	2.4M	0.8M
Robotics	0.4M	0.2M	4	1.6M	0.8M
BNP2	4.0M	1.0M	1	4.0M	1.0M
TOTAL				**20.3M**	**2.8M**

Draft

Sticker shock? We remain positive...

Taking full advantage of the MBA upgrade will require considerable effort and a bit of luck to obtain the funding needed to be successful. Our membership is excited by the possibilities and that's always a good way to start.

The future of LS-CAT Discussion

Response to proposed amendment to extend the LS-CAT contract to 2022:

Institution	Status	Shares
Northwestern	Supportive	1
University of Michigan	Supportive	1
Van Andel	Supportive	1
Michigan State	Supportive at ½ share	0.5
Wisconsin	Supportive	0.5
Illinois	Unknown	
Vanderbilt	Unknown	
Wayne State	Possible	“0.5”

LS-CAT Status

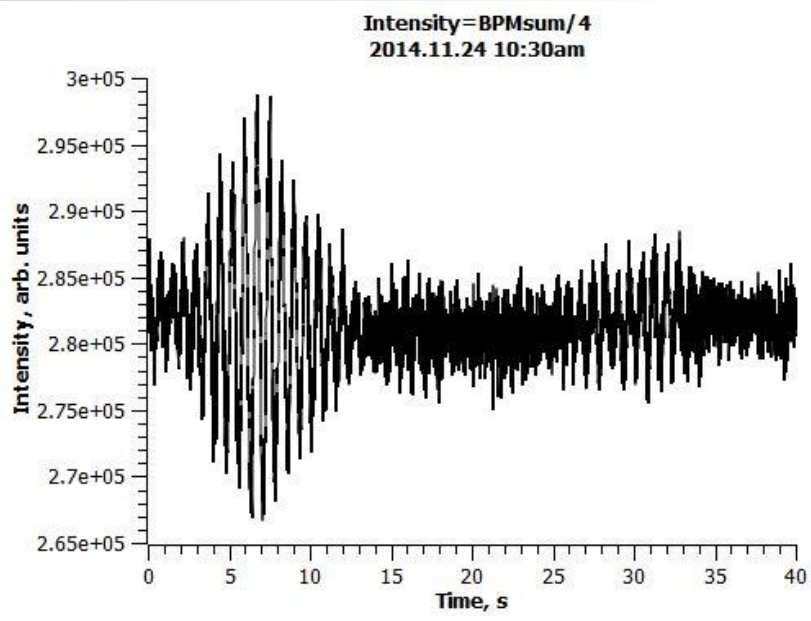
21-ID-D Monochromator Problems

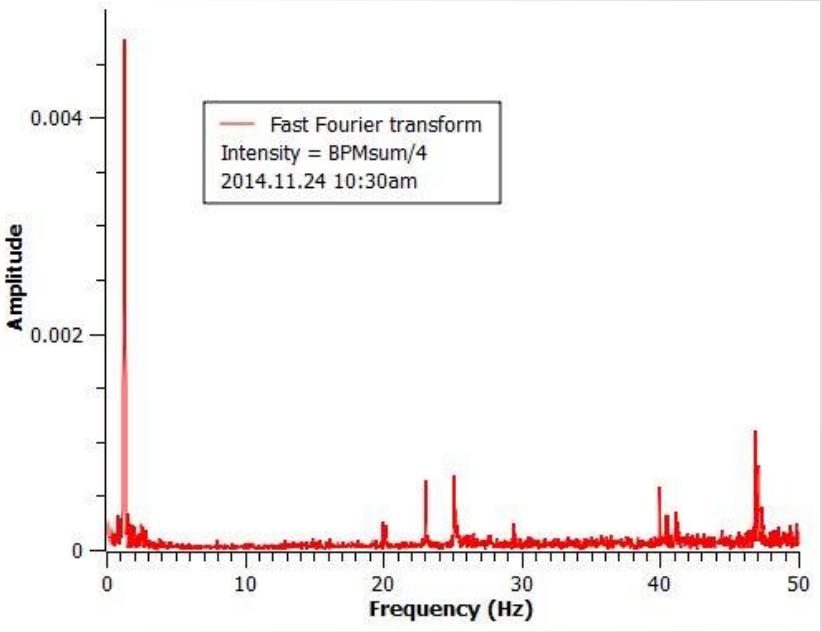
Every year in September we warm up the mono and send the cryopump off for preventative maintenance.

This year when we came back up in October there was pronounced beam motion to the extent that collecting data with the vertical focusing mirror was impossible

We spent considerable time ruling out various causes and now have a fairly long list of things that are not problems.

21-ID-D Monochromator Problems





21-ID-D Monochromator Problems

Things that are not the cause:

1. The APS
2. Either of the mirrors
3. LN2 flow rate
4. LN2 cryopump
5. Water flow
6. Various fans
7. Loose second crystal
8. Loose first crystal
9. Any of the slits
10. Any obvious new vibration source

Things we'll be trying this down period:

1. Evaluate possible causes with the tank warm and open
2. Rebuild the mono (with repolished first and second crystals)
3. Replumb LN2 lines in the mono tank to reduce inertial moments of the piping and to decouple first and second crystals

The “Smart” Magnet



The MD-2 diffractometer uses a circuit in its magnet power supply to detect changes in inductance indicating the presence of a cap on the diffractometer.

LS-CAT has used this circuit to control the CATS robots to tell if a manually mount sample is present or if a dismount failed. HOWEVER, this approach has become unreliable over the past couple of years. (Why? We do not know, but it has been the biggest single source of robot problems at LS-CAT.)

Last year we ordered and installed replacement circuits and replacement magnets. The new ones were even worse and the manufacturer has been unable to get them to work.

Time for new tactics.

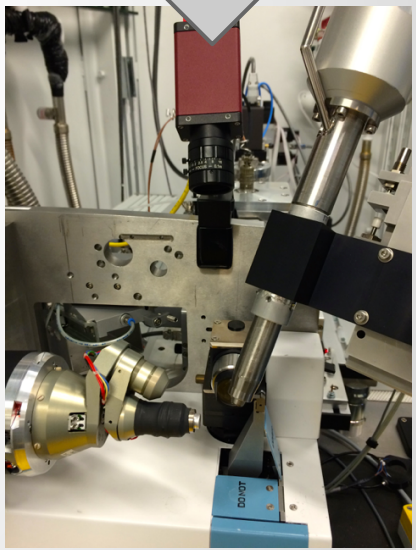
Our solution: Provide a machine vision approach to cap detection.

Software Development

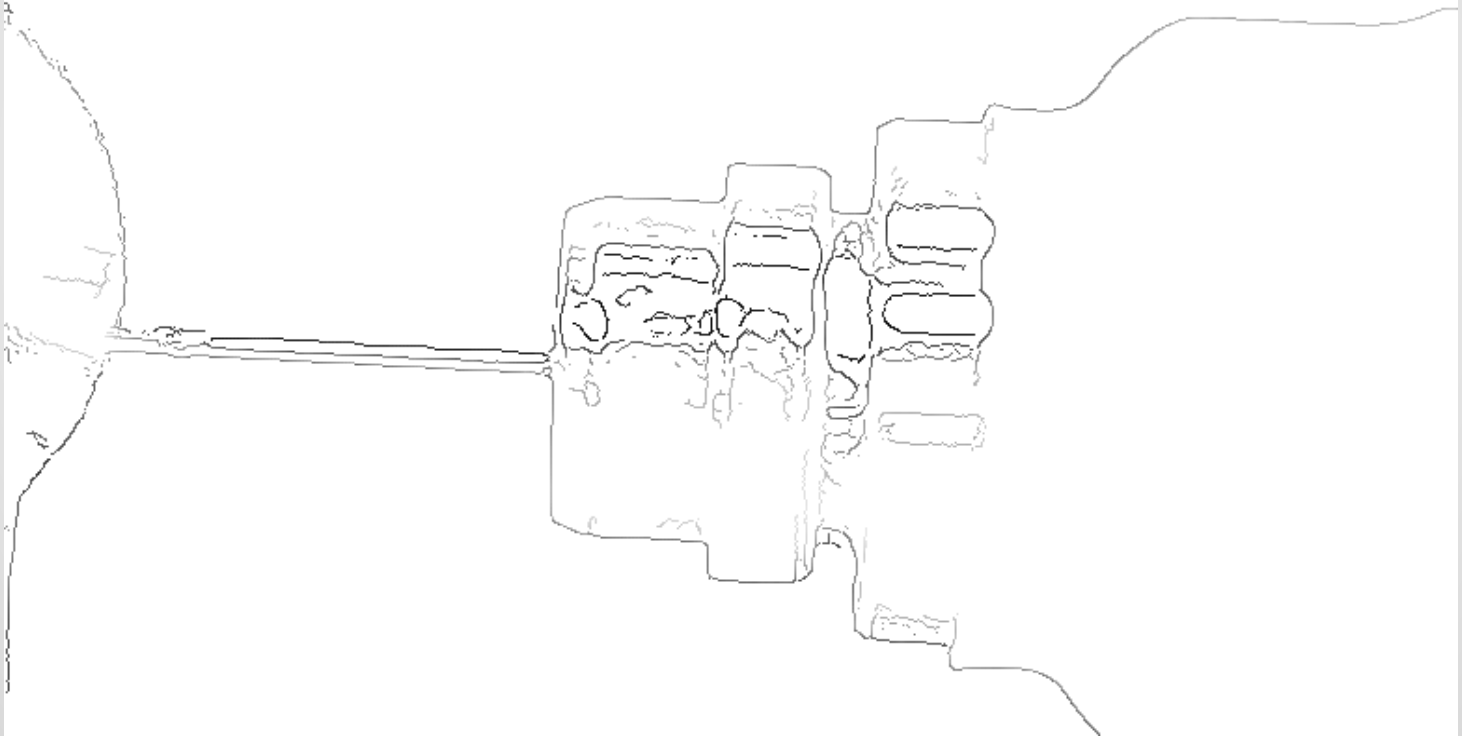
- Installed new cameras to observe the spindle
  - Uses EPICS areaDetector (also used for other projects below)
  - Installed edge detection (also needed for new remote features)
- New Remote Interface
  - New Data Collection Modes
- Digital MD-2 Cameras
- Eiger Integration

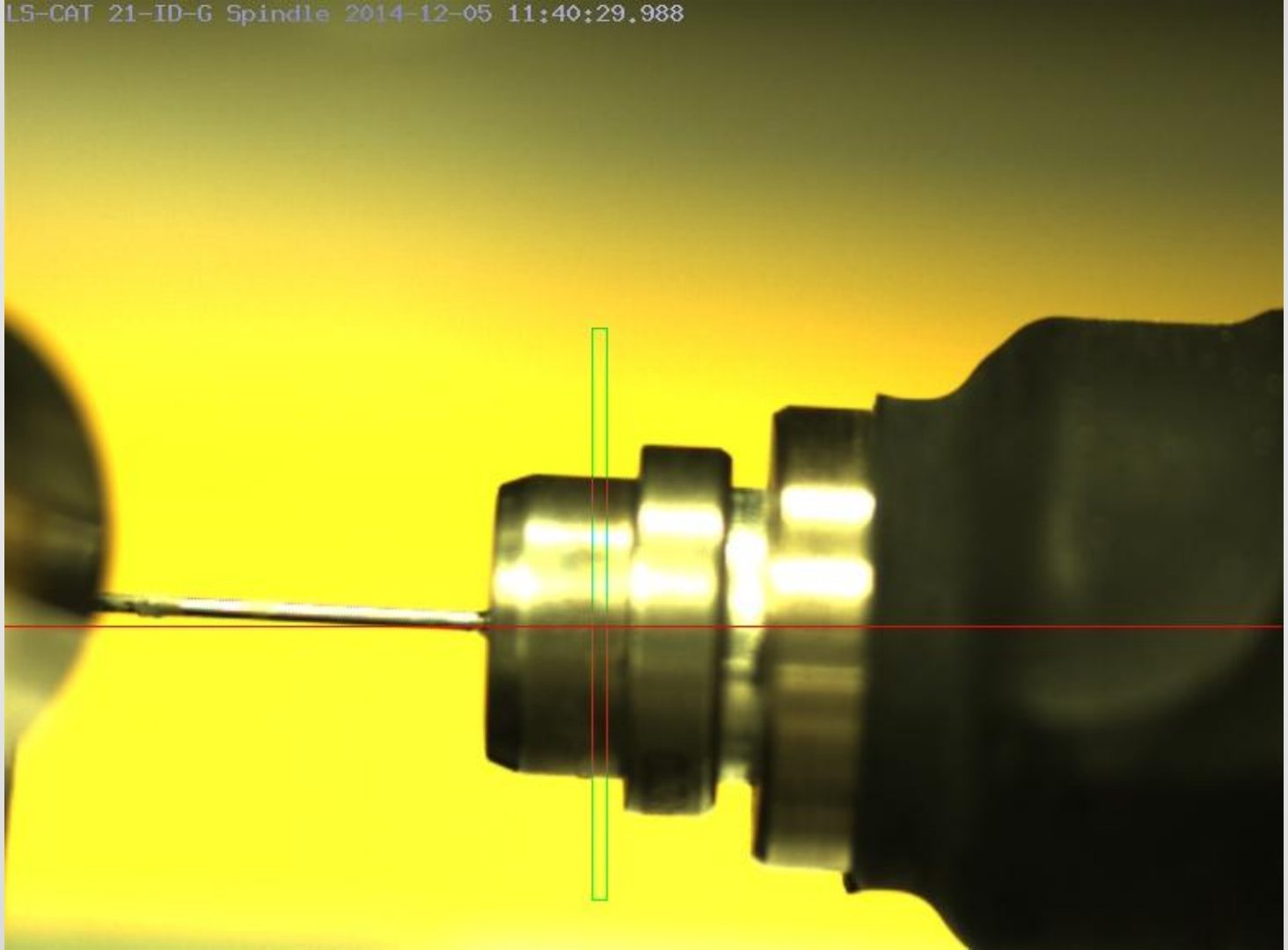
Starting next run!

Improve performance especially for local users. Point of no return for old software.

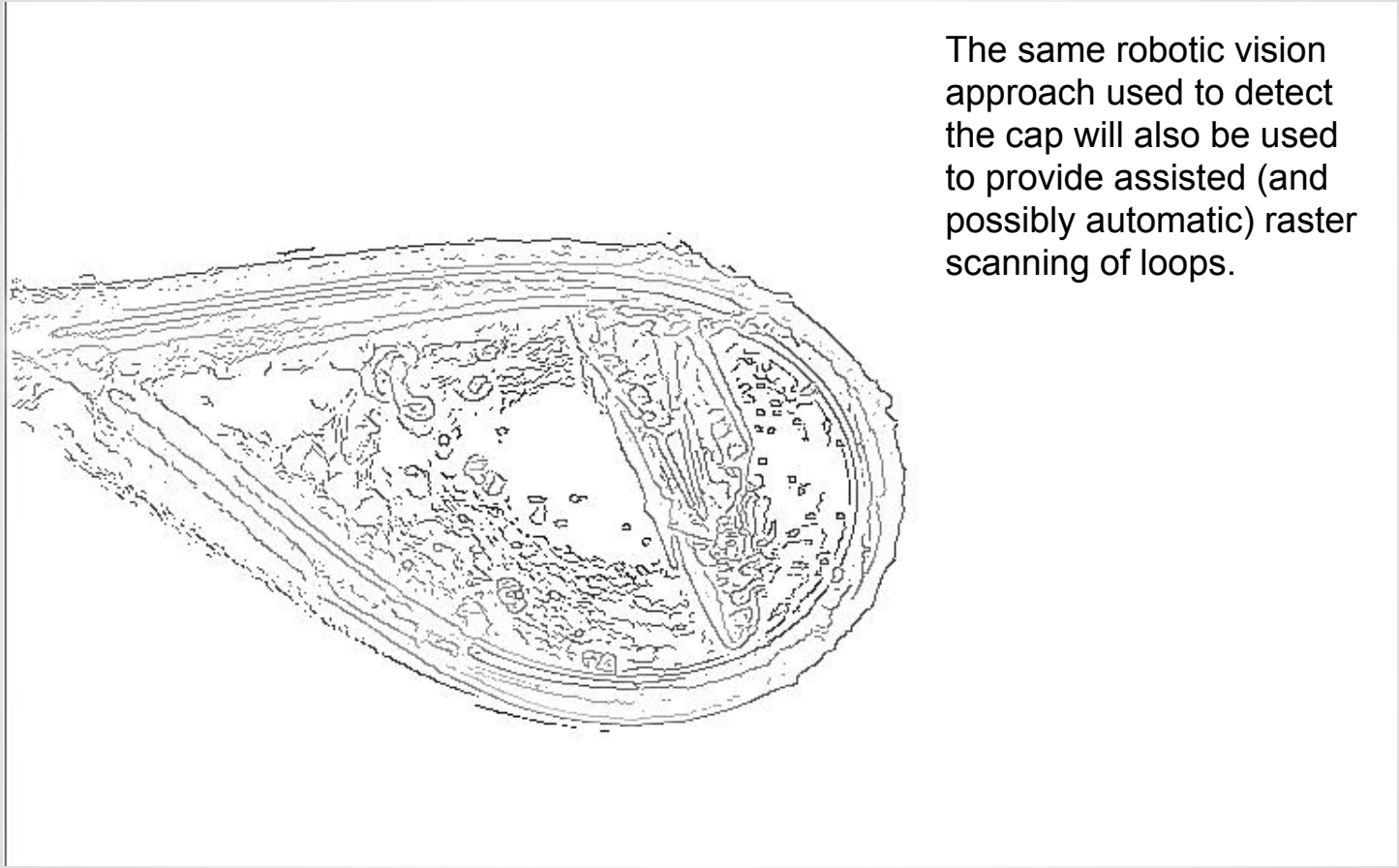







LS-CAT 21-ID-G Spindle 2014-12-05 11:40:29.988





The same robotic vision approach used to detect the cap will also be used to provide assisted (and possibly automatic) raster scanning of loops.

Next Meeting

Possible Agenda Items:

- APS Upgrade Office Update ⇐ They need to hear what your needs are.
- The Budget
- Election for the next board chair

May 11 - 14, 2015: APS Users Meeting, just before or just after, not during.